

# Business Requirements Document

## Connect Broadband Customer Churn & Call Center Analytics Dashboard

### Project Overview

Connect Broadband is an internet service provider serving residential and business customers. Customer retention and service quality are critical factors that directly impact revenue, customer satisfaction, and business growth.

The company requires a centralized reporting solution to monitor customer churn trends and evaluate customer support performance. The objective is to provide business stakeholders with actionable insights that help reduce customer attrition, improve service quality, and optimize call center operations.

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### Business Objectives

The dashboard should enable stakeholders to:

- Monitor customer churn trends.
  - Identify factors contributing to customer churn.
  - Quantify revenue loss caused by customer attrition.
  - Analyze churn across customer segments.
  - Evaluate call center performance and efficiency.
  - Monitor issue resolution rates.
  - Measure customer satisfaction levels.
  - Improve customer retention strategies.
  - Support data-driven operational decisions.
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### Stakeholders

#### Primary Users

- Customer Retention Team
- Customer Support Managers
- Operations Managers
- Business Analysts
- Senior Management

## Secondary Users

- Customer Service Agents
  - Marketing Team
  - Revenue Management Team
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# Dashboard Requirements

## Page 1: Customer Churn Analysis

### Purpose

Provide insights into customer churn behavior, customer demographics, and revenue loss associated with churned customers.

### Key KPIs

- Total Customers
- Churn Count
- Churn Percentage
- Monthly Revenue Loss
- Young Customer Churn Count
- Senior Customer Churn Count

### Visual Requirements

#### Customer Churn Distribution

Display total customers by churn status:

- Churned
- Active

#### Churn by Tenure

Analyze churn across tenure groups:

- 0–10 Months
- 10–20 Months
- 20–30 Months
- 30–40 Months
- 40–50 Months
- 50–60 Months
- 60–70 Months
- 70+ Months

### **Churn by Customer Age Group**

Analyze churn among:

- Young Customers
- Senior Customers

### **Churn by Payment Method**

Identify payment methods associated with higher churn rates.

### **Churn by Contact Type**

Evaluate the relationship between customer contact methods and churn behavior.

### **Monthly Charges Analysis**

Analyze monthly charges by tenure and churn status.

### **Churn by Internet Service**

Identify internet service categories with the highest churn rates.

### **Filters**

- Month
  - Customer Segment
  - Payment Method
  - Internet Service
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## **Page 2: Call Center Analytics**

### **Purpose**

Provide visibility into customer support performance, issue resolution effectiveness, and agent productivity.

### **Key KPIs**

- Average Speed of Answer
- Total Calls Answered
- Average Talk Duration
- Total Resolved Calls
- Average Calls Per Day
- Total Unresolved Calls

## Visual Requirements

### Resolution Status Overview

Display:

- Resolved Calls %
- Unresolved Calls %

### Resolution by Topic

Analyze resolution rates across support topics.

### Call Volume Analysis

Track:

- Calls Answered
- Resolved Calls
- Unresolved Calls

by support topic.

### Agent Performance Analysis

Include:

- Agent Name
- Calls Answered
- Total Resolved
- Resolution Percentage
- Customer Satisfaction Rating
- Average Talk Duration

### Customer Satisfaction Analysis

Measure and compare agent performance using customer satisfaction scores.

### Filters

- Month
- Call Topic
- Agent Name

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## Business Questions Addressed

### Customer Churn Analysis

- What percentage of customers are churning?

- Which customer segments are at the highest risk of churn?
- Does customer tenure impact churn behavior?
- Which payment methods are associated with higher churn rates?
- Which internet services experience the highest customer attrition?
- What is the monthly revenue impact of churn?

## Call Center Analytics

- What percentage of customer issues are resolved successfully?
  - Which support topics generate the highest call volumes?
  - How quickly are customer calls answered?
  - Which agents achieve the highest resolution rates?
  - How satisfied are customers with support services?
  - What factors contribute to unresolved issues?
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## Data Sources

### Customer Churn Dataset

Contains:

- Customer ID
- Customer Demographics
- Tenure
- Internet Service Type
- Payment Method
- Monthly Charges
- Contact Type
- Churn Status

### Call Center Dataset

Contains:

- Agent Information
  - Call Topic
  - Call Date
  - Resolution Status
  - Talk Duration
  - Satisfaction Rating
  - Calls Answered
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## Data Model Requirements

The solution should utilize a scalable data model supporting:

## Fact Tables

- Customer Churn Data
- Call Center Activity Data

## Dimension Tables

- Date
- Customer
- Internet Service
- Payment Method
- Agent
- Call Topic

The model should support fast filtering, drill-down analysis, and accurate KPI calculations.

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## Success Criteria

The dashboard will be considered successful if it enables stakeholders to:

- Identify key churn drivers.
  - Quantify revenue loss due to churn.
  - Improve customer retention efforts.
  - Increase issue resolution rates.
  - Monitor agent performance effectively.
  - Improve customer satisfaction.
  - Reduce manual reporting efforts.
  - Support faster business decision-making.
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## Tools & Technologies

- Power BI
  - Power Query
  - DAX
  - Excel
  - Data Modeling
  - Data Visualization
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## Expected Business Impact

The dashboard will provide a single source of truth for customer retention and support operations. By combining churn analytics with call center performance metrics, stakeholders can proactively address customer concerns, reduce customer attrition, improve service quality, and enhance overall customer experience. The solution will support data-driven decision-making and contribute to improved customer loyalty and business growth.